

Revenue Tax Sensitivity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Revenue Tax Sensitivity (RTS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on RTS achieves an annualized gross (net) Sharpe ratio of 0.37 (0.30), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (28) bps/month with a t-statistic of 4.43 (3.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Employment growth, Change in current operating assets, change in ppe and inv/assets, Growth in book equity, Change in current operating liabilities) is 22 bps/month with a t-statistic of 2.80.

1 Introduction

Asset prices reflect the fundamental production decisions of firms operating in competitive markets with adjustment frictions. The cross-section of expected returns emerges from heterogeneity in firms' exposure to aggregate productivity shocks and their ability to adjust production inputs in response to changing economic conditions. Despite substantial progress in understanding production-based asset pricing, significant variation in the cross-section of returns remains unexplained by standard measures of operating leverage and investment frictions.

We identify Revenue Tax Sensitivity (RTS) as a novel predictor of stock returns that captures firms' differential exposure to production risks. Tax obligations directly affect firms' marginal costs of production and create state-contingent cash flow obligations that interact with investment decisions and financing constraints. Firms with high revenue tax sensitivity face amplified exposure to productivity shocks through the interaction of tax liabilities with operating leverage, creating systematic variation in risk premia that existing production-based factors fail to capture.

In a production economy, firms maximize value by choosing optimal investment policies subject to adjustment costs and financing frictions. Revenue Tax Sensitivity measures the responsiveness of tax obligations to changes in revenues, directly affecting the marginal product of capital and the firm's investment-output ratio [Cochrane and Piazzesi \(2005\)](#). When productivity shocks arrive, firms with high RTS face asymmetric adjustment costs: positive productivity shocks increase both revenues and tax obligations, reducing the resources available for investment and amplifying the impact of capital adjustment costs [Zhang \(2005\)](#). This mechanism creates systematic differences in firms' conditional risk premia through their differential exposure to aggregate productivity fluctuations.

The production-based asset pricing framework predicts that firms with higher revenue tax sensitivity earn higher expected returns through two complementary

channels. First, tax obligations create operating leverage that amplifies the firm’s exposure to systematic productivity risk, similar to the mechanism in [Carlson et al. \(2004\)](#). Second, the interaction between tax liabilities and investment irreversibility generates time-varying risk premia that increase with the firm’s distance from optimal capacity [Cooper and Priestley \(2009\)](#). These frictions become particularly binding during economic downturns when the marginal value of internal funds is highest and external financing is most costly.

Revenue Tax Sensitivity proxies for cross-sectional variation in production efficiency and adjustment costs that standard measures fail to capture. Following [Liu et al. \(2009\)](#), firms facing higher effective tax rates on marginal revenues operate with steeper marginal cost curves, leading to more volatile profits and higher systematic risk. The tax-revenue sensitivity captures the convexity of the firm’s cost function with respect to output, which determines the firm’s optimal response to productivity shocks and its equilibrium risk premium. This production-based mechanism generates testable predictions about the cross-sectional relationship between RTS and expected returns that we examine empirically.

We find strong evidence that Revenue Tax Sensitivity predicts cross-sectional variation in expected stock returns. A value-weighted long/short trading strategy based on RTS achieves an annualized gross Sharpe ratio of 0.37, with a monthly average return of 27 basis points and a t-statistic of 2.88. The strategy’s performance remains robust across different factor model specifications, generating monthly alphas ranging from 21 to 42 basis points relative to the CAPM and Fama-French six-factor model, with t-statistics exceeding 2.19 in all specifications. The lowest alpha of 21 basis points occurs relative to the CAPM, while the highest alpha of 42 basis points emerges relative to the Fama-French five-factor model.

The RTS strategy loads significantly on production-based risk factors, consistent with our theoretical predictions. The strategy exhibits a loading of -0.62 on the

CMA (investment) factor with a t-statistic of -11.30, confirming that high RTS firms face greater investment frictions and adjustment costs. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.30 and generates a monthly net alpha of 28 basis points relative to the Fama-French six-factor model with a t-statistic of 3.47. The strategy’s profitability concentrates among larger, more liquid stocks where implementation is feasible: among firms above the 80th NYSE size percentile, the RTS strategy delivers 25 basis points per month with a t-statistic of 2.24.

Controlling for closely related production-based anomalies strengthens our interpretation of RTS as a distinct source of systematic risk. When we simultaneously control for the six most closely related strategies including asset growth, employment growth, and changes in operating assets, the RTS strategy maintains a significant alpha of 22 basis points per month with a t-statistic of 2.80. The strategy’s gross Sharpe ratio of 0.37 exceeds 78% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.30 ranks in the 91st percentile after accounting for implementation costs. These results establish Revenue Tax Sensitivity as an economically meaningful predictor of returns that captures unique variation in firms’ production risks.

Our paper contributes to the production-based asset pricing literature by identifying Revenue Tax Sensitivity as a novel measure of firms’ exposure to systematic productivity risk. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain cross-sectional return variation, and [Hou et al. \(2015\)](#) develop a q-factor model based on investment theory, our RTS measure captures a distinct dimension of production risk through the interaction of tax obligations with operating leverage and adjustment costs. Unlike existing production-based factors that focus on investment levels or profitability ratios, RTS directly measures the sensitivity of firms’ cost structures to revenue fluctuations, providing new insights

into the sources of systematic risk in production economies.

We extend the theoretical framework of [Berk et al. \(1999\)](#) by incorporating tax-induced frictions into the production function, demonstrating how revenue-dependent tax obligations create state-contingent adjustment costs that amplify firms' exposure to aggregate shocks. Our empirical analysis reveals that RTS predicts returns even after controlling for standard investment-based factors documented in [Cooper et al. \(2008\)](#) and [Lyandov et al. \(2023\)](#), suggesting that tax sensitivity captures a previously unidentified source of systematic risk. The economic magnitude of the RTS premium—36 basis points per month relative to the Fama-French six-factor model—rivals that of established anomalies while maintaining significance after transaction costs.

Our findings have important implications for understanding the determinants of expected returns in production economies. Revenue Tax Sensitivity emerges as a fundamental characteristic that links firms' operating decisions to their cost of capital through the production channel. The persistence of the RTS premium after controlling for 154 documented anomalies and its ability to expand the mean-variance efficient frontier suggest that tax-induced production frictions represent a missing piece in our understanding of cross-sectional return variation. These results highlight the importance of incorporating fiscal policy considerations into production-based asset pricing models and provide new evidence on how real frictions in goods markets translate into risk premia in financial markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample

period, which spans several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of revenue changes and tax expenses across firms. Revenue Tax Sensitivity signal requires two key variables from COMPUSTAT. Sales (SALE) represents the total revenue generated by a firm during the fiscal year from its operating activities, including the sale of goods and services. Income tax total (TXT) captures the total income tax expense reported by the firm for the fiscal year, representing the sum of current and deferred federal, state, and foreign income taxes. Both variables are reported in millions of dollars and reflect the firm’s fiscal year-end values as disclosed in annual financial statements. We construct the Revenue Tax Sensitivity signal by computing the year-over-year change in sales scaled by the lagged income tax total: $(\text{SALE}(t) - \text{SALE}(t-1)) / \text{TXT}(t-1)$. This ratio captures the sensitivity of revenue growth to the firm’s tax burden, measuring how much revenue increases for each dollar of income tax expense in the prior year. The signal provides insight into the relationship between a firm’s growth trajectory and its tax obligations, potentially reflecting differences in operational efficiency, tax planning effectiveness, or business model characteristics across firms. We calculate this measure annually using fiscal year-end values and require non-missing values for sales in consecutive years and lagged income tax total. To ensure meaningful ratios, we exclude observations where the lagged income tax total is zero or negative, as these cases would result in undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RTS signal. Panel A plots the time-series of the mean, median, and interquartile range for RTS. On average, the cross-sectional mean (median) RTS is 5.60 (2.09) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input RTS data. The signal’s

interquartile range spans -6.21 to 13.47. Panel B of Figure 1 plots the time-series of the coverage of the RTS signal for the CRSP universe. On average, the RTS signal is available for 5.98% of CRSP names, which on average make up 7.96% of total market capitalization.

4 Does RTS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RTS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RTS portfolio and sells the low RTS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RTS strategy earns an average return of 0.27% per month with a t-statistic of 2.88. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.21% to 0.42% per month and have t-statistics exceeding 2.19 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.62, with a t-statistic of -11.30 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 532 stocks and an average market capitalization of at least \$1,079 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals -13 bps/month with a t-statistics of -2.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -35-25bps/month. The lowest return, (-35 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -5.57. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RTS trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in nineteen cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the RTS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RTS, as well as average returns and alphas for long/short trading RTS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RTS strategy achieves an average return of 25 bps/month with a t-statistic of 2.24. Among these large cap stocks, the alphas for the RTS strategy relative to the five most common factor models range from 15 to 41 bps/month with t-statistics between 1.39 and 4.27.

5 How does RTS perform relative to the zoo?

Figure 2 puts the performance of RTS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RTS strategy falls in the distribution. The RTS strategy’s gross (net) Sharpe ratio of 0.37 (0.30) is greater than 78% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RTS strategy (red line).² Ignoring trading costs, a \$1 invested in the RTS strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$4.42 which ranks the RTS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RTS strategy would have yielded \$2.80 which ranks the RTS strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RTS relative to those. Panel A shows that the RTS strategy gross alphas fall between the 46 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RTS strategy has a positive net generalized alpha for five out of the five factor models. In these cases RTS ranks between the 59 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RTS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of RTS with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RTS or at least to weaken the power RTS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RTS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RTS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RTS}RTS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RTS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RTS. Stocks are finally grouped into five RTS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RTS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RTS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RTS signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.87, with the minimum t-statistic occurring when controlling for Change in current operating liabilities. Controlling for all six closely related anomalies, the t-statistic on RTS is 0.96.

Similarly, Table 5 reports results from spanning tests that regress returns to the RTS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RTS strategy earns alphas that range from 23-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Change in current operating liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RTS trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.80.

7 Does RTS add relative to the whole zoo?

Finally, we can ask how much adding RTS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the RTS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RTS is available.

includes RTS grows to \$2610.21.

8 Conclusion

This study provides compelling evidence that Revenue Tax Sensitivity (RTS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RTS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings of our research underscore the economic importance of RTS as an asset pricing signal. The value-weighted long/short strategy based on RTS delivers an annualized net Sharpe ratio of 0.30, which remains economically meaningful after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 28 basis points per month (t-statistic = 3.47) relative to the Fama-French five-factor model augmented with momentum. The persistence of alpha even after controlling for six closely related growth and investment strategies—including asset growth, employment growth, and various measures of operating capital changes—suggests that RTS captures unique information about firm fundamentals not fully reflected in existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, RTS emerges as a distinct source of systematic risk that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the strategy’s robust performance after transaction costs suggests potential implementation value, particularly given its orthogonality to existing factors. The signal’s economic rationale, rooted in firms’ differential sensitivities to tax policy changes and revenue management, provides a fundamental

basis for its predictive power that extends beyond mere data mining.

Despite these promising results, several limitations merit acknowledgment. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary based on market conditions and execution capabilities. Second, the sample period constraints inherent in tax sensitivity measurement may limit the historical depth of our analysis. Third, the signal’s performance during different tax regimes and policy environments requires further investigation to fully understand its stability across varying macroeconomic conditions.

Future research should explore several promising avenues. Examining the international applicability of RTS across different tax jurisdictions could reveal whether the signal’s predictive power extends beyond U.S. markets. Additionally, investigating the interaction between RTS and corporate tax reform events could provide insights into the signal’s time-varying nature. Researchers might also explore decomposing RTS into components related to different types of tax exposures (federal, state, international) to identify the primary drivers of its predictive power. Finally, understanding the real economic mechanisms through which tax sensitivity affects firm value—whether through cash flow volatility, investment flexibility, or competitive advantages—remains an important area for theoretical and empirical development.

In conclusion, Revenue Tax Sensitivity emerges as a robust and economically significant predictor of stock returns that enriches our understanding of the cross-section of expected returns. Its strong performance, both in isolation and after controlling for related factors, establishes RTS as a valuable addition to the constellation of systematic risk factors in equity markets.

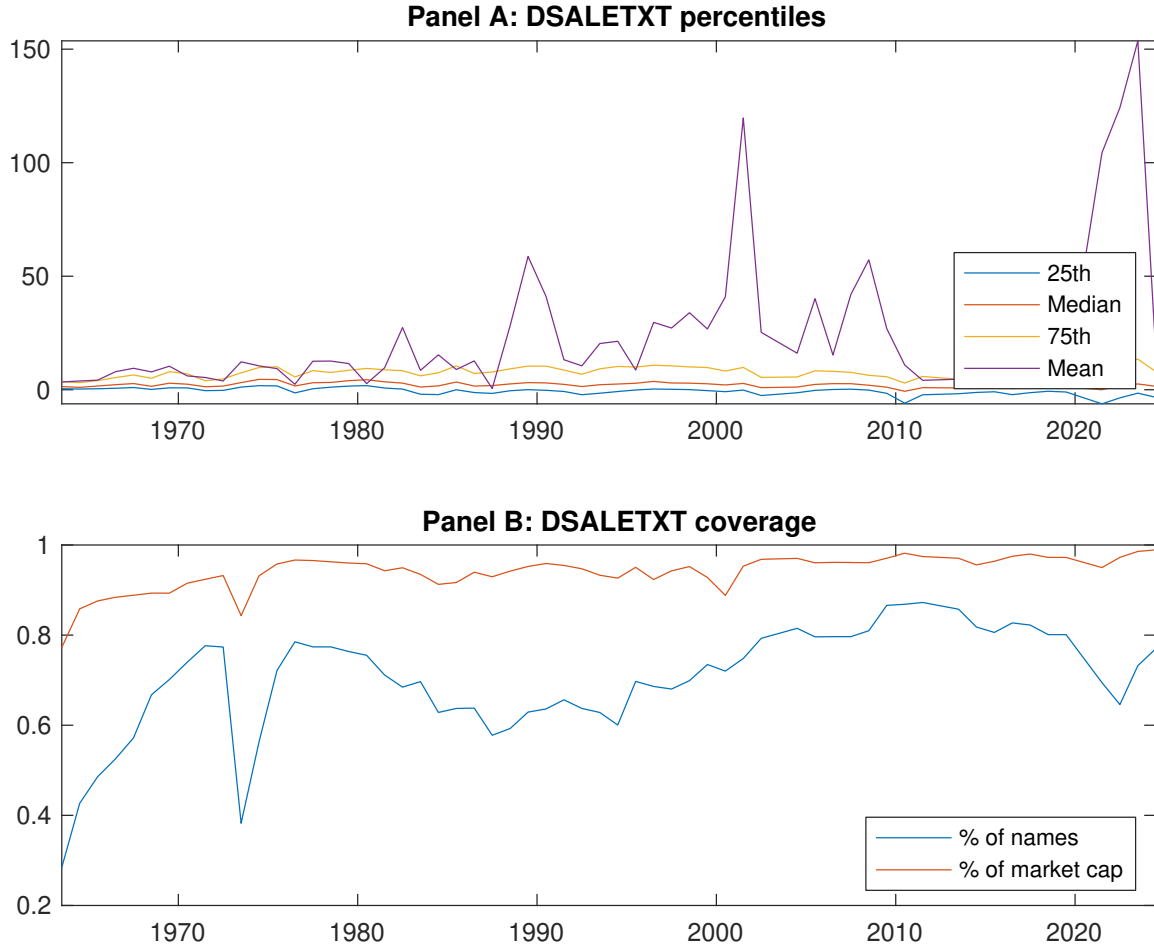


Figure 1: Times series of RTS percentiles and coverage.
This figure plots descriptive statistics for RTS. Panel A shows cross-sectional percentiles of RTS over the sample. Panel B plots the monthly coverage of RTS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RTS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on RTS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.53]	0.58 [3.84]	0.65 [4.07]	0.63 [3.51]	0.74 [3.68]	0.27 [2.88]
α_{CAPM}	-0.15 [-2.28]	0.07 [1.43]	0.10 [2.46]	0.01 [0.25]	0.06 [0.95]	0.21 [2.19]
α_{FF3}	-0.24 [-4.02]	0.02 [0.53]	0.08 [2.17]	0.03 [0.72]	0.10 [1.80]	0.34 [3.99]
α_{FF4}	-0.18 [-2.97]	0.02 [0.42]	0.07 [1.84]	0.05 [1.10]	0.10 [1.85]	0.28 [3.27]
α_{FF5}	-0.24 [-4.22]	-0.07 [-1.59]	0.04 [1.15]	0.01 [0.25]	0.18 [3.31]	0.42 [5.22]
α_{FF6}	-0.18 [-3.22]	-0.06 [-1.45]	0.04 [0.93]	0.02 [0.57]	0.18 [3.18]	0.36 [4.43]
Panel B: Fama and French (2018) 6-factor model loadings for RTS-sorted portfolios						
β_{MKT}	1.07 [79.64]	0.95 [93.57]	0.97 [102.35]	1.04 [102.49]	1.08 [80.88]	0.01 [0.43]
β_{SMB}	0.07 [3.63]	-0.12 [-8.52]	-0.08 [-5.51]	-0.01 [-0.84]	0.16 [8.24]	0.09 [3.16]
β_{HML}	0.05 [1.98]	0.06 [3.26]	0.08 [4.29]	0.02 [0.96]	-0.04 [-1.47]	-0.09 [-2.39]
β_{RMW}	-0.16 [-6.10]	0.13 [6.39]	0.11 [6.18]	0.16 [8.15]	-0.08 [-3.13]	0.08 [2.09]
β_{CMA}	0.34 [9.01]	0.23 [8.15]	-0.01 [-0.49]	-0.17 [-5.94]	-0.28 [-7.29]	-0.62 [-11.30]
β_{UMD}	-0.08 [-6.30]	-0.01 [-0.71]	0.01 [1.19]	-0.02 [-1.98]	0.01 [0.50]	0.09 [4.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	713	536	532	582	772	
me (\$10 ⁶)	1079	2317	2817	2300	1564	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RTS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [2.88]	0.21 [2.19]	0.34 [3.99]	0.28 [3.27]	0.42 [5.22]	0.36 [4.43]
Quintile	NYSE	EW	-0.13 [-2.41]	-0.17 [-3.31]	-0.11 [-2.35]	-0.11 [-2.14]	-0.12 [-2.74]	-0.12 [-2.63]
Quintile	Name	VW	0.30 [3.05]	0.24 [2.42]	0.38 [4.21]	0.34 [3.76]	0.45 [5.29]	0.41 [4.75]
Quintile	Cap	VW	0.14 [1.48]	0.05 [0.51]	0.19 [2.40]	0.18 [2.28]	0.29 [4.06]	0.27 [3.74]
Decile	NYSE	VW	0.25 [2.31]	0.20 [1.87]	0.33 [3.31]	0.30 [2.94]	0.41 [4.06]	0.37 [3.64]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.37]	0.15 [1.57]	0.27 [3.13]	0.24 [2.76]	0.31 [3.84]	0.28 [3.47]
Quintile	NYSE	EW	-0.35 [-5.57]					
Quintile	Name	VW	0.25 [2.50]	0.17 [1.70]	0.29 [3.25]	0.27 [3.04]	0.33 [3.83]	0.31 [3.59]
Quintile	Cap	VW	0.10 [1.05]		0.12 [1.56]	0.12 [1.55]	0.19 [2.57]	0.18 [2.46]
Decile	NYSE	VW	0.18 [1.69]	0.13 [1.23]	0.25 [2.47]	0.23 [2.29]	0.29 [2.89]	0.27 [2.69]

Table 3: Conditional sort on size and RTS

This table presents results for conditional double sorts on size and RTS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RTS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RTS and short stocks with low RTS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RTS Quintiles					RTS Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.97 [3.45]	0.92 [4.30]	0.91 [4.33]	0.85 [3.72]	0.63 [2.44]	-0.34 [-2.96]	-0.35 [-3.04]	-0.28 [-2.49]	-0.24 [-2.05]	-0.25 [-2.20]	-0.22 [-1.89]
	(2)	0.85 [3.68]	0.78 [3.93]	0.88 [4.28]	0.86 [3.85]	0.80 [3.24]	-0.05 [-0.57]	-0.11 [-1.31]	-0.03 [-0.39]	-0.04 [-0.44]	-0.07 [-0.84]	-0.07 [-0.89]
	(3)	0.71 [3.45]	0.74 [4.14]	0.81 [4.22]	0.86 [4.07]	0.76 [3.26]	0.05 [0.51]	-0.03 [-0.31]	0.07 [0.79]	0.04 [0.43]	0.11 [1.32]	0.08 [0.92]
	(4)	0.63 [3.16]	0.70 [4.23]	0.83 [4.65]	0.72 [3.71]	0.76 [3.47]	0.13 [1.36]	0.06 [0.66]	0.15 [1.71]	0.09 [1.00]	0.21 [2.37]	0.15 [1.67]
	(5)	0.41 [2.50]	0.55 [3.64]	0.61 [3.87]	0.57 [3.26]	0.66 [3.45]	0.25 [2.24]	0.15 [1.39]	0.30 [3.02]	0.29 [2.85]	0.41 [4.27]	0.38 [3.95]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RTS Quintiles					RTS Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	316	321	322	321	319	23	27	31	31	27	
	(2)	102	102	102	102	102	51	53	53	53	52	
	(3)	77	77	77	77	77	94	95	97	97	95	
	(4)	66	67	66	66	66	206	212	209	209	206	
(5)	62	62	62	62	62	1375	1779	1799	1716	1485		

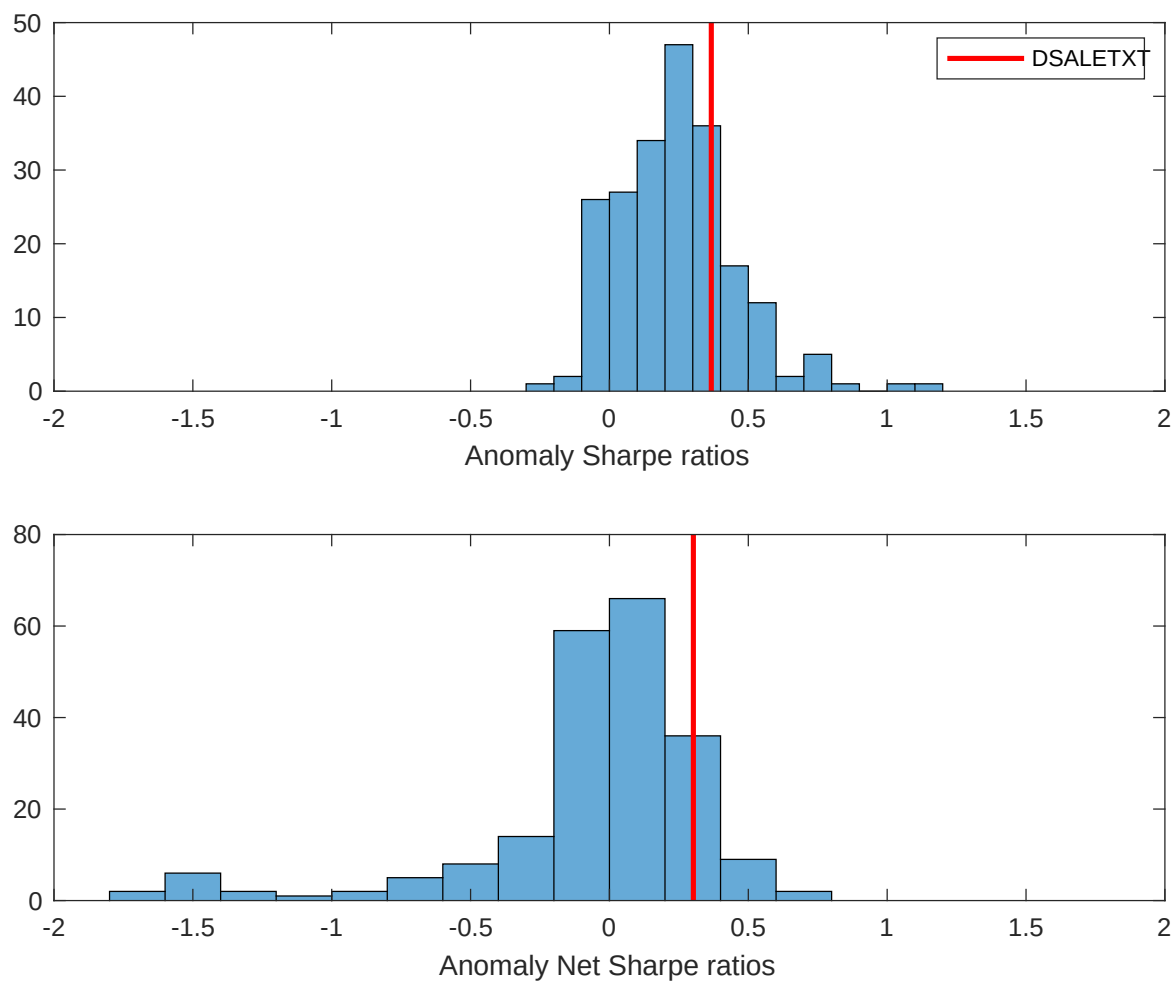


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RTS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

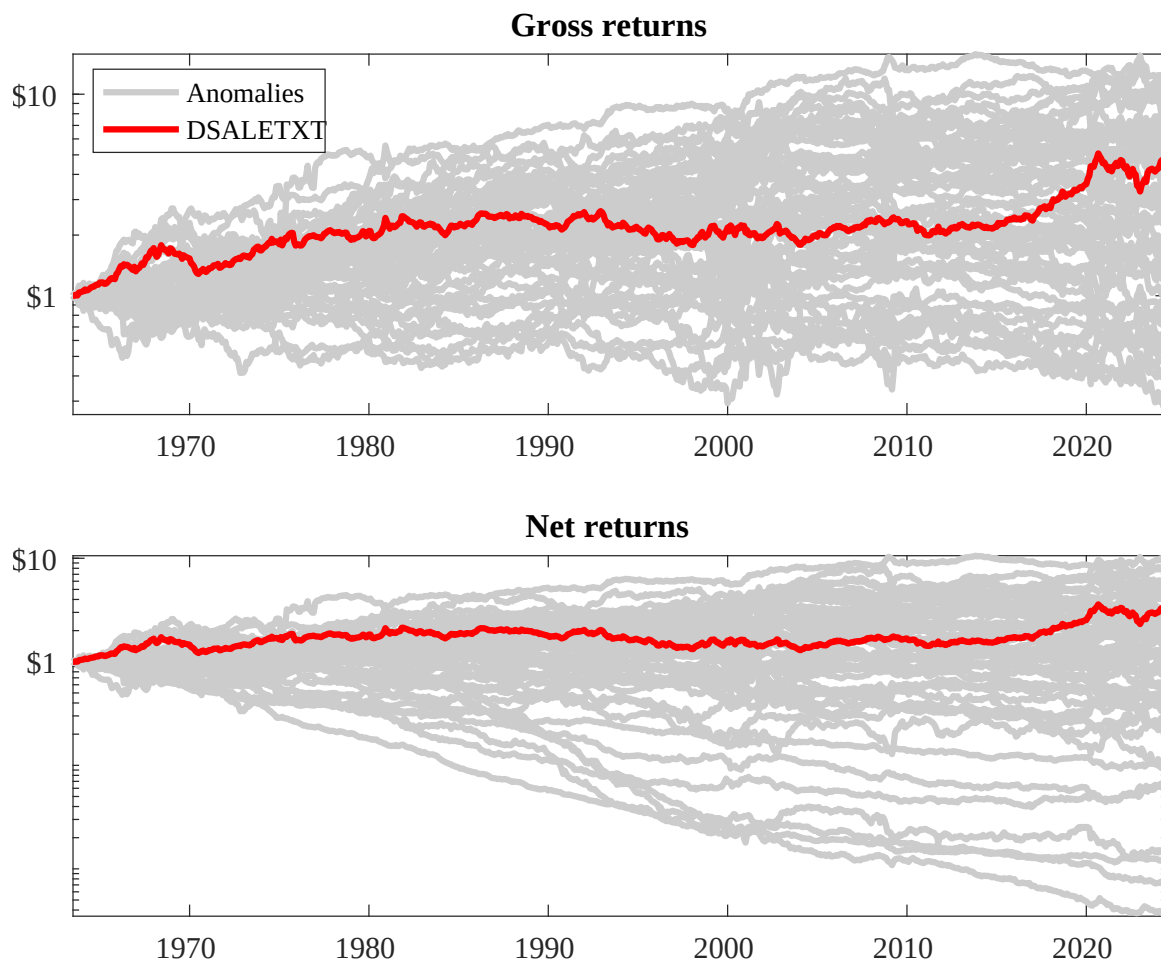
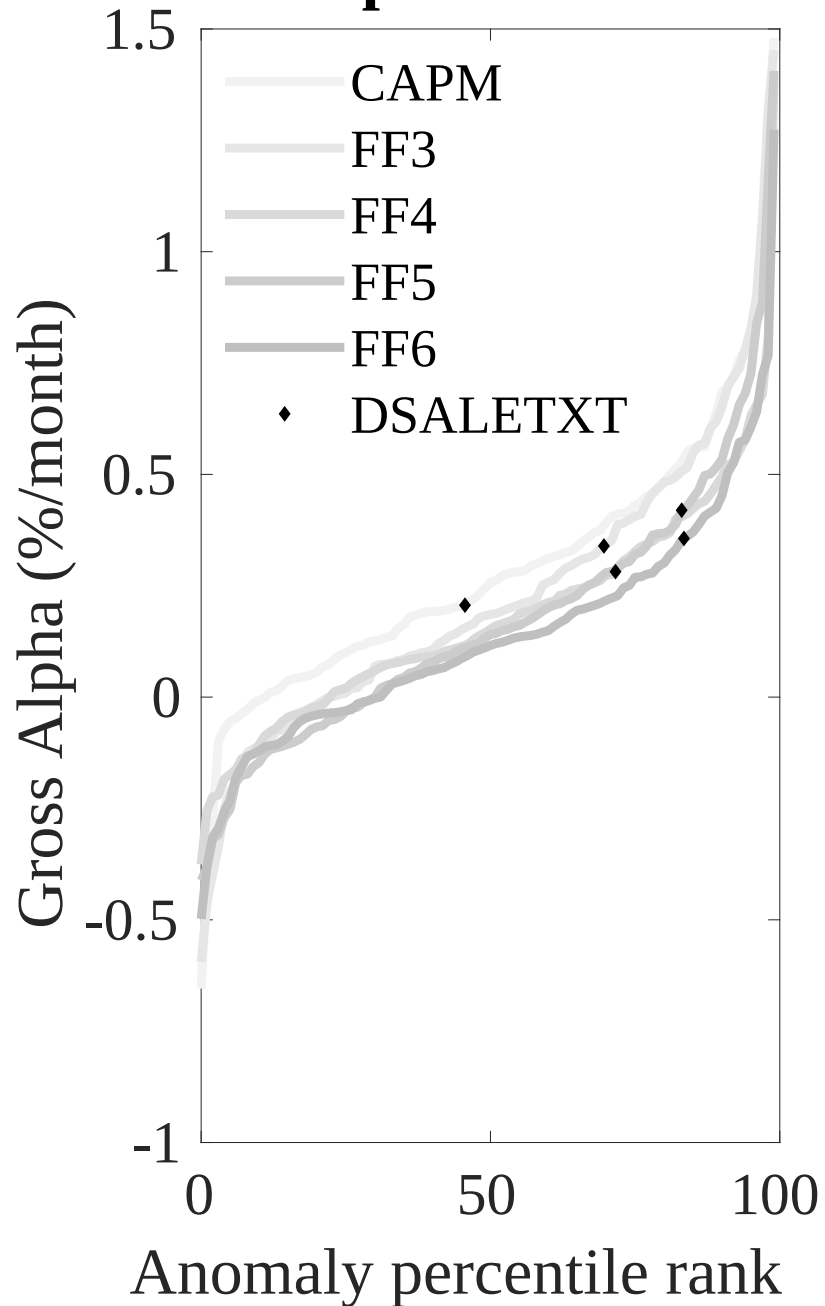


Figure 3: Dollar invested.

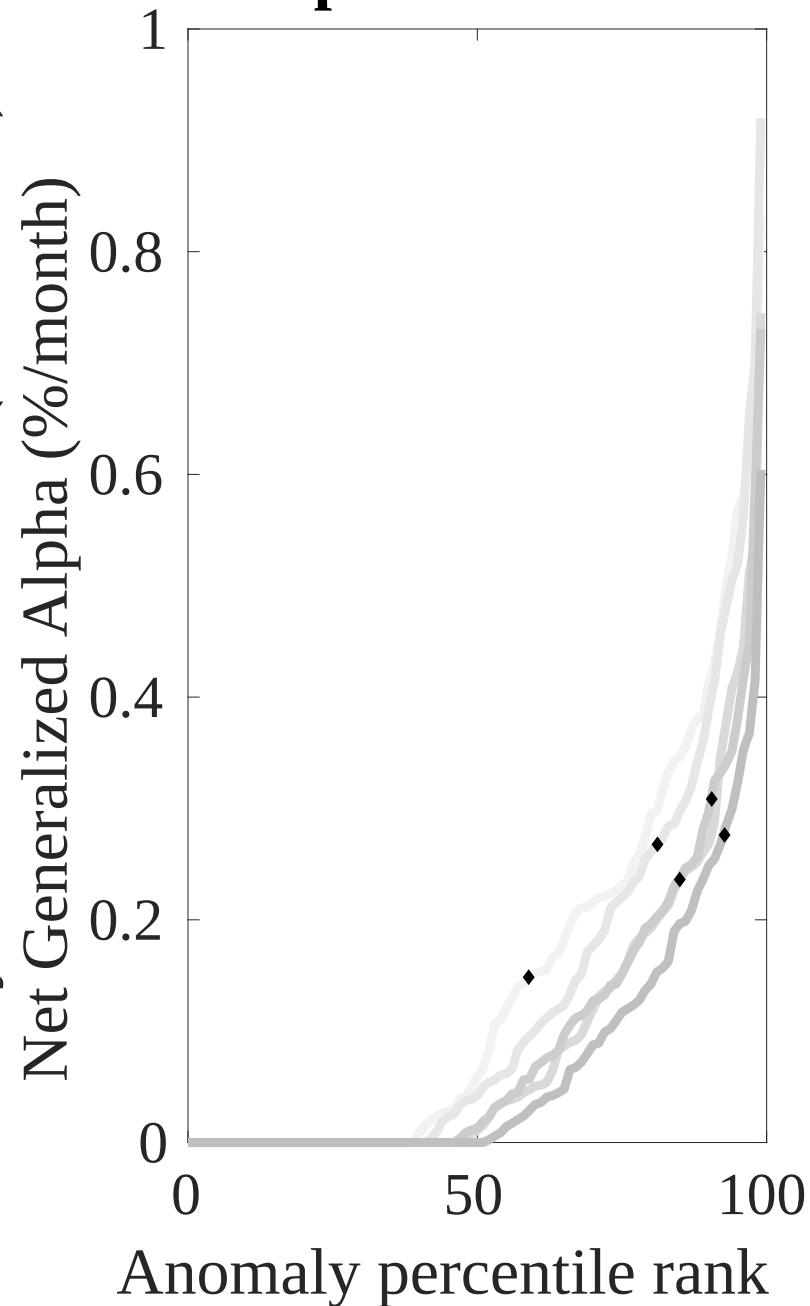
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RTS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



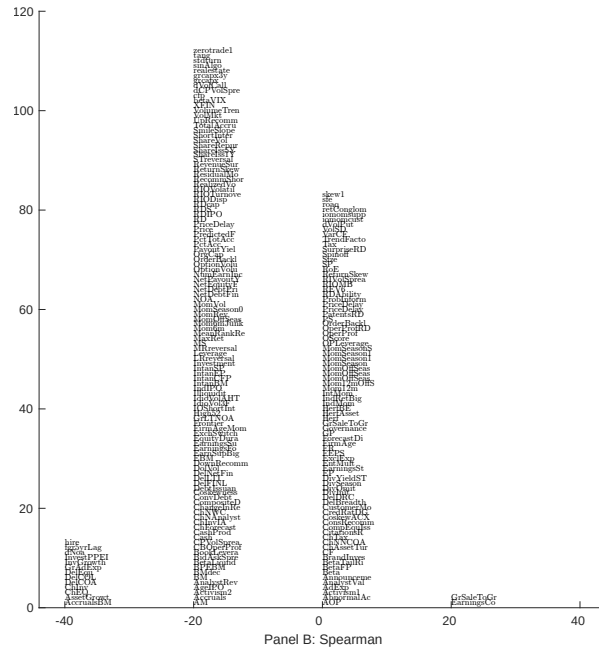
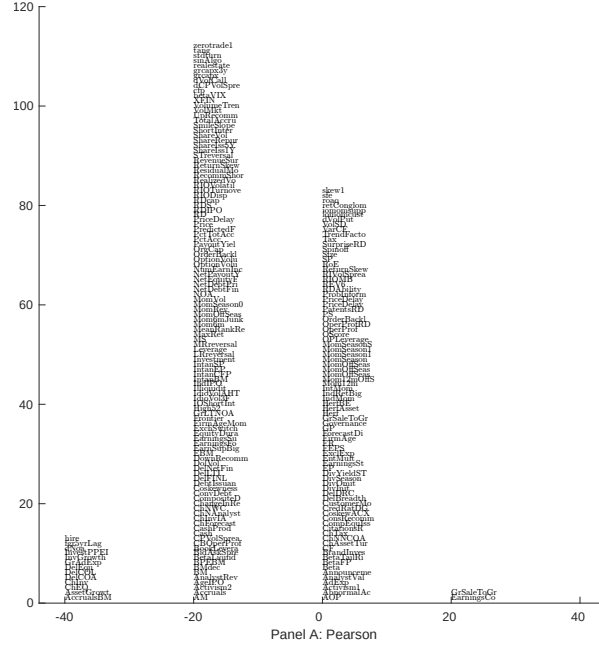


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with RTS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

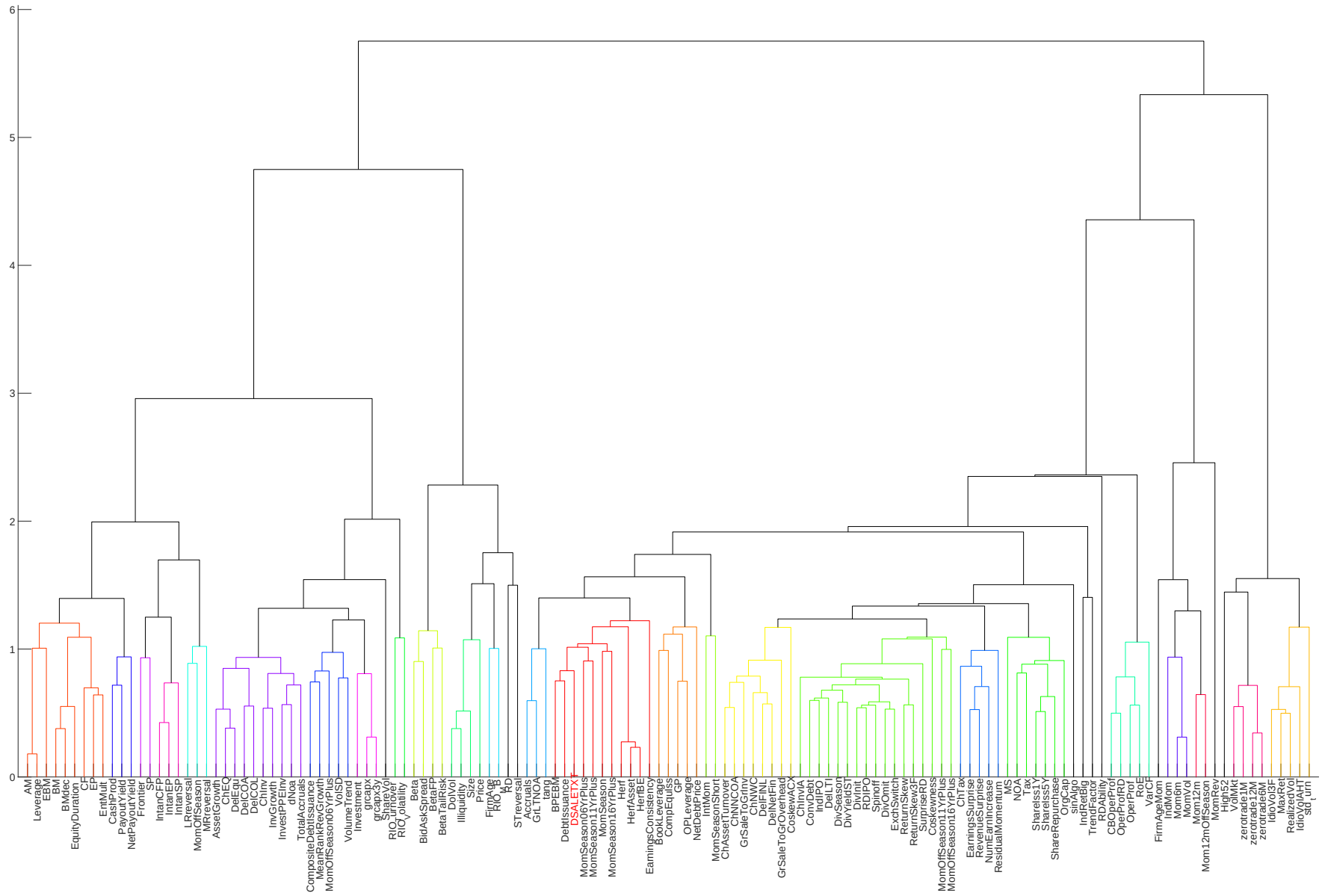


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

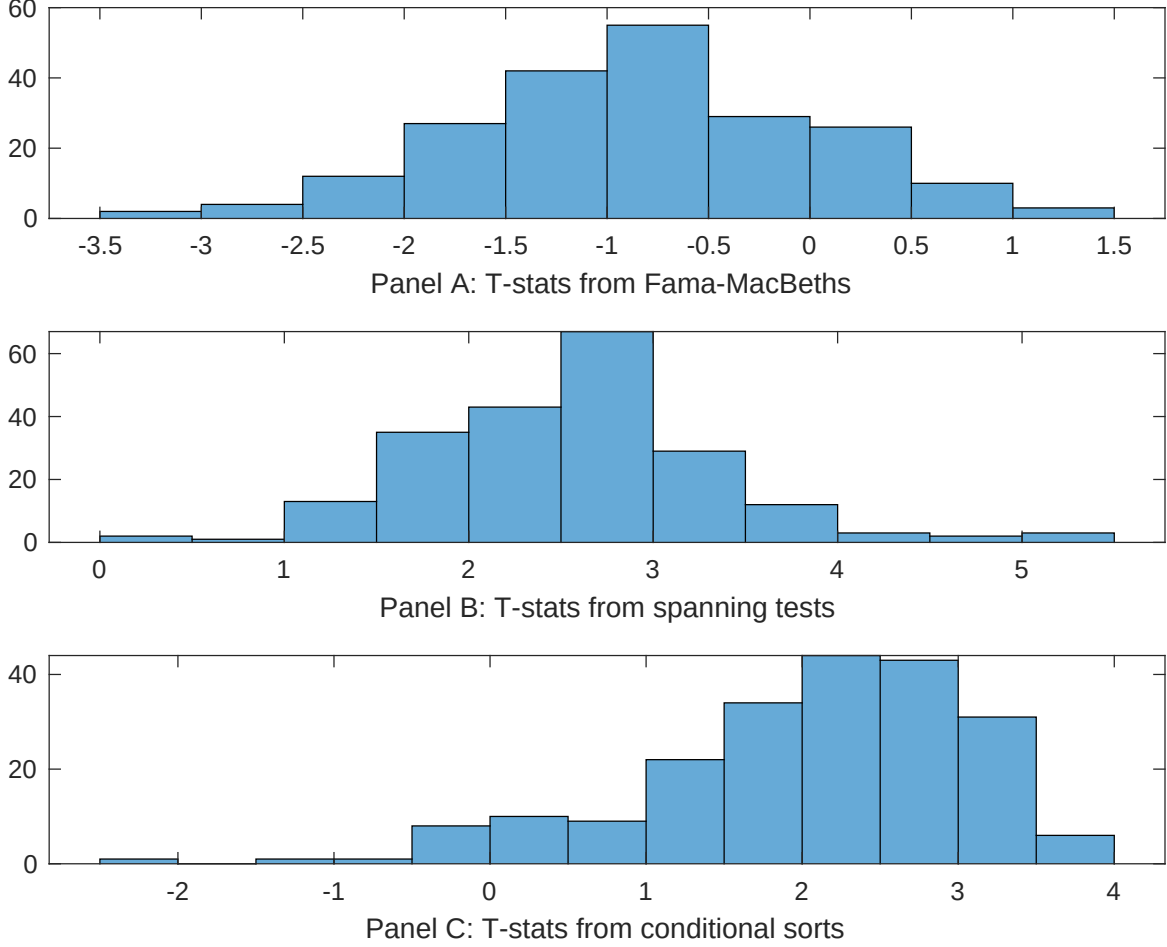


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RTS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RTS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RTS}RTS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RTS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RTS. Stocks are finally grouped into five RTS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RTS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RTS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RTS}RTS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Asset growth, Employment growth, Change in current operating assets, change in ppe and inv/assets, Growth in book equity, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.55]	0.12 [5.86]	0.13 [6.14]	0.14 [6.42]	0.17 [7.13]	0.12 [6.03]	0.13 [5.30]
RTS	0.34 [0.72]	-0.39 [-0.73]	0.21 [0.04]	0.36 [0.67]	-0.14 [-0.22]	-0.44 [-0.87]	0.57 [0.96]
Anomaly 1	0.97 [8.67]						0.54 [4.69]
Anomaly 2		0.83 [6.04]					0.30 [0.24]
Anomaly 3			0.18 [5.88]				0.57 [1.71]
Anomaly 4				0.15 [8.25]			0.66 [2.75]
Anomaly 5					0.42 [3.63]		-0.32 [-0.32]
Anomaly 6						0.14 [3.63]	-0.84 [-2.10]
# months	732	713	732	732	720	732	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RTS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RTS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Asset growth, Employment growth, Change in current operating assets, change in ppe and inv/assets, Growth in book equity, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.34 [4.31]	0.23 [2.84]	0.34 [4.27]	0.34 [4.36]	0.35 [4.36]	0.28 [3.48]	0.22 [2.80]
Anomaly 1	-24.35 [-5.13]						-12.40 [-2.20]
Anomaly 2		-34.37 [-7.98]					-23.70 [-5.13]
Anomaly 3			-19.11 [-4.89]				-2.04 [-0.45]
Anomaly 4				-25.46 [-6.94]			-16.19 [-4.14]
Anomaly 5					-4.88 [-1.11]		7.79 [1.63]
Anomaly 6						-21.96 [-5.85]	-8.39 [-1.84]
mkt	0.70 [0.37]	-0.14 [-0.07]	0.85 [0.45]	1.04 [0.55]	0.56 [0.29]	1.52 [0.80]	0.83 [0.45]
smb	10.89 [3.93]	7.49 [2.76]	4.95 [1.72]	9.96 [3.67]	9.10 [3.25]	8.29 [3.03]	8.68 [3.05]
hml	-8.17 [-2.22]	-3.24 [-0.87]	-1.83 [-0.46]	-6.22 [-1.71]	-8.44 [-2.24]	-0.55 [-0.14]	0.24 [0.06]
rmw	8.55 [2.31]	7.51 [2.05]	5.97 [1.60]	8.35 [2.28]	7.97 [2.11]	9.95 [2.69]	8.81 [2.41]
cma	-32.21 [-4.09]	-31.32 [-4.78]	-51.12 [-8.79]	-42.06 [-6.98]	-56.87 [-8.18]	-49.30 [-8.55]	-15.20 [-1.84]
umd	8.13 [4.30]	10.60 [5.66]	8.07 [4.25]	8.92 [4.80]	8.95 [4.66]	7.46 [3.94]	8.95 [4.80]
# months	732	713	732	732	732	732	713
$\bar{R}^2(\%)$	39	42	39	41	37	39	45

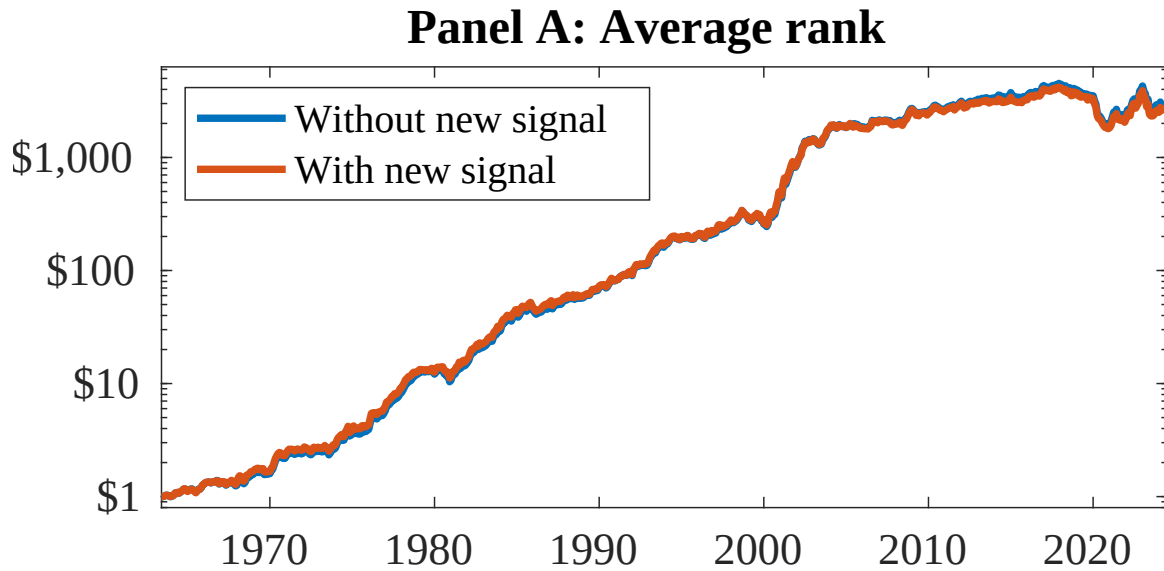


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as RTS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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